



Research article

Privacy and cybersecurity in tension: Unpacking EU public opinion on online child protection measures

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ABSTRACT

In this study, we examine the tension between privacy concerns and cybersecurity measures in the context of online child sexual abuse material (CSAM) detection within the European Union (EU). Using data from the Flash Eurobarometer 532, we investigate public opinion on proposed EU legislation aimed at combating online CSAM. Our objective is to uncover nuanced perspectives on privacy and child protection that may be obscured by aggregate statistics. We employ K-means cluster analysis to identify distinct groups within the population based on their attitudes towards online privacy, child safety, and support for CSAM detection measures. Our results reveal three main opinion clusters in the EU27 population: Privacy-focused (22%), Action-focused (30%), and CSAM-focused (48%). These groups differ significantly in their prioritization of privacy rights versus child protection and their support for various detection methods. We find that while overall support for CSAM detection measures is high across all scenarios, including those with significant privacy implications, cluster analysis reveals important variations in public opinion. These variations are associated with demographic factors such as age, gender, and education, as well as geographic location within the EU. Our findings illustrate the trade-offs between privacy and child protection, and highlight the importance of considering the diverse public perspectives when developing policies to address online CSAM. This study contributes to a more nuanced understanding of public opinion on the balance between privacy rights and cybersecurity measures in the digital age.

1. Introduction

The prevalence of child sexual abuse material online is both extensive and growing. Estimates suggest that millions of images and videos depicting sexual abuse of minors circulate worldwide, often shared via social media, peer-to-peer networks, messaging platforms, and dark web forums [1,2]. High-profile incidents, such as the exposure of large-scale child sexual abuse material (CSAM) distribution networks or viral circulation of abuse images, highlight the transnational and persistent nature of this problem [3]. Emerging technologies, including generative AI, can further amplify harmful content, creating new channels for exploitation and complicating detection efforts [4].

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Exposure to sexual abuse has profound and long-lasting effects on child victims. Individuals whose images or videos are circulated online often experience trauma, anxiety, depression, and social withdrawal, compounded by repeated re-victimization as content continues to circulate [5–7]. Beyond the immediate psychological harm, victims may face social stigmatization, difficulties in forming trusting relationships, and ongoing fear for personal safety. The continued availability of abuse material perpetuates harm to survivors and enables the grooming of additional children and the normalization of exploitative behaviors [7]. Recognizing these severe consequences is crucial to understanding the social imperative for legislative and technological interventions aimed at protecting children.

The debate over compulsory detection of CSAM by online service providers centers on balancing child protection with privacy and security concerns. Proponents argue that mandatory detection could reduce CSAM spread, deter offenders, and accelerate victim identification. They note that hash-matching technologies are already used to detect previously identified CSAM [8,9]. However, detection approaches raise privacy issues, especially for encrypted communications, and could lead to false positives [10,11]. They also impose significant implementation costs on companies. Thus, opponents emphasize the importance of preserving user privacy and data protection. They argue that compulsory detection could compromise encryption integrity and create vulnerabilities exploitable by malicious actors, including corporate and state actors [11]. However, this stance may allow more CSAM to remain undetected and slow investigations.

The controversy extends across various platform types, including public communications, email, chat, and end-to-end encrypted services. These service types create distinct legal, security, and privacy challenges [10,11]. The debate also questions the appropriate division of responsibility between private companies and government agencies. Key issues include the effectiveness of automated detection versus human review, especially in a context of generative AI-driven amplification of CSAM, the potential for government overreach, and the risk of driving offenders to less detectable platforms [12–15].

The European Commission presented a controversial legislative proposal on May 11, 2022, aimed at preventing and combating child sexual abuse online [8,16]. This proposal seeks to address the growing challenge of CSAM in the digital world, as current voluntary detection and reporting measures by companies have proven insufficient. The proposal would require relevant service providers to assess and mitigate the risk that their services are misused for online child sexual abuse. Where preventive measures are insufficient, competent authorities could seek targeted detection orders. The proposal also sets out reporting, removal, and access-blocking obligations and provides for the establishment of an EU Centre on Child Sexual Abuse [8].

However, this proposal has sparked intense debate and criticism, particularly regarding its potential impact on privacy rights. Critics argue that the obligations imposed on online service providers, including messaging services and web-based email, could create a massive new surveillance system [16–18]. The controversy surrounding the proposal has pitted child protection advocates against digital rights groups. Supporters, including survivors of online child abuse, argue that the regulation is crucial for protecting children and combating the rising tide of CSAM online. For example, Rhiannon, a survivor and advocate, emphasizes the importance of stronger regulation to clamp down on sexual predators and prevent the re-traumatization of victims whose abuse images continue to circulate online [18]. On the other hand, privacy advocates and some academic experts warn about the potential risks of the proposed technologies. Professor Bart Preneel, a cryptography expert, raises concerns about the accuracy of detection methods and the risk of false positives, which could lead to wrongful accusations of innocent users [18].

Critics argue that applying the proposed detection obligations to encrypted services could require or incentivize client-side scanning (CSS) [11,18]. Child-protection advocates argue that technological detection remains necessary for combating online abuse, whereas security and privacy experts warn that applying detection obligations to encrypted services could weaken practical confidentiality and expand surveillance capabilities [11].

Our study is based on the secondary analysis of the European Commission Flash Eurobarometer 532 dataset, which aims to assess the public opinion in European Union (EU27) concerning this legislative proposal. We aim to examine how European Union citizens navigate the tension between online privacy rights and support for measures to detect and prevent online CSAM. Moving beyond the aggregate findings reported in the Eurobarometer survey, we seek to uncover underlying patterns of public opinion that may be obscured by high levels of support for child protection measures. Specifically, we investigate whether distinct opinion profiles exist within the EU27 population that reveal different ways of balancing privacy concerns with support for CSAM detection measures.

This study builds on our earlier analysis of the same Flash Eurobarometer 532 dataset [19], which examined Europeans' privacy calculus and security concerns in online CSAM detection using multinomial and country-level regression models. The present study is distinct in terms of analytical strategy and contribution. Whereas the earlier, variable-centered study modeled the individual- and country-level predictors of privacy and security attitudes, here we adopt a person-centered approach. We use factor analysis to construct summative attitude indices and K-means cluster analysis to identify discrete opinion profiles within the EU27 population. This clustering approach surfaces latent heterogeneity that mean-level or regression-based analyses cannot reveal, and it reframes the privacy-protection tension as a question of how distinct publics, rather than isolated predictors, organize their views.

The remainder of this paper is organized as follows. In the next section, State of the art, we review the literature on the technological, legal, and ethical controversies surrounding online CSAM detection and summarize the findings of the Flash Eurobarometer 532 report. In the Materials and Methods section, we describe the dataset and the analytical strategy. The Results section reports the descriptive findings, the three identified opinion clusters, their socio-demographic profiles, and their country-level distribution. Finally, in the Discussion and Conclusion sections, we interpret the findings, consider their implications for policy and future research, and acknowledge the limitations of the study.

1.1. State of the art

In recent years, the proliferation of online CSAM has emerged as a critical issue at the intersection of child protection, technology,

and privacy rights. To better understand the challenges involved in addressing online CSAM, we discuss recent studies that shed light on the technological, legal, and ethical dimensions of CSAM detection and prevention, as well as public perceptions and expectations regarding online child protection measures.

Lee et al. [9] conducted a review of CSAM detection, focusing on policy frameworks, distribution channels, and detection applications. Their research highlighted the global nature of CSAM as an online crime, emphasizing the need for international cooperation in legislation and enforcement. The study revealed that weak or inconsistent laws across countries create loopholes that perpetrators can exploit. In this context, cybercrime investigations often run into jurisdictional challenges due to the borderless nature of the internet, which complicates enforcement, cooperation, and extradition across legal systems. Another area of concern is digital evidence. Ensuring admissibility in court requires integrity, proper chain of custody, authentication, and consistent legal frameworks, yet many jurisdictions lag behind in maintaining standardized rules and oversight [20].

Researchers noted a positive trend in global legislation, with more countries implementing comprehensive CSAM laws, though gaps remain [21]. The challenges posed by rapidly evolving technology can outpace existing laws and create new avenues for CSAM distribution. Distribution methods for CSAM are diverse, including peer-to-peer networks, the darknet, web search engines, mobile devices, and social media platforms [9]. Each channel presents unique challenges for detection and enforcement. The rise of social media is particularly concerning, as it provides new platforms for offenders to contact potential victims and distribute CSAM. Lee et al. [9] highlighted the psychological and emotional trauma experienced by victims of CSAM, noting that the continuous circulation of abuse material online leads to perpetual re-victimization. The study also touches on the potential negative impacts of CSAM investigations on law enforcement personnel, emphasizing the need for automated detection to protect workers from trauma.

Razi et al. [22] presented a systematic literature review of computational approaches for online sexual risk detection, employing a human-centered lens. The findings revealed that most research has concentrated on detecting sexual grooming, which accounts for 75% of studies, while less attention has been given to sex trafficking and sexual harassment/abuse. Notably, the majority of approaches aim to detect risks after they have occurred, with 93% focusing on post-incident detection rather than prevention. A significant issue highlighted in the review is the lack of ecological validity in the datasets used. Many studies rely heavily on public datasets like Perverted Justice, which do not accurately reflect real-world interactions. Furthermore, ground truth annotations frequently depend on third-party annotators without relevant expertise, rather than incorporating perspectives from victims or subject matter experts. The authors argue for a more human-centered approach to this research. They suggest using more ecologically valid datasets that reflect real user interactions, involving domain experts and victims in data annotation, and incorporating contextual and behavioral features into models. They also advocate for conducting user studies and real-world evaluations, developing deployable systems and interventions rather than just algorithms, and carefully considering the ethical implications and potential misuse of these technologies.

Singh and Nambiar's [23] systematic literature review explored the potential of artificial intelligence (AI) in preventing and detecting online child sexual abuse (OCSA). The authors conducted a comprehensive analysis of 35 papers to identify AI-driven interventions, their limitations, and future implications. The study highlighted three main themes: current AI trends in OCSA prevention, algorithm evaluation, and recommendations for advancing AI tools. While the review suggested that AI technologies such as machine learning, pattern recognition, and chatbots show promise in identifying and preventing child abuse online, it also revealed a lack of conclusive evidence regarding their effectiveness. The authors emphasized the need for more extensive, applied research to better understand and improve AI-based interventions in combating OCSA.

The study conducted by Sanchez et al. [24] explored the value that practitioners in CSAM investigations assign to current forensic tools, artificial intelligence, filtering technologies, and safer presentation methods. The researchers employed a mixed-methods approach, utilizing a survey with 106 participants from various law enforcement agencies and organizations. The study revealed several key findings. Practitioners predominantly use commercial tools for processing CSAM, with a preference for filtering technologies over safe-viewing technologies. The main concerns regarding current tools include speed, accuracy, and reporting capabilities, which significantly impact workflow efficiency. Participants expressed a strong desire for improved workflows, emphasizing the need for adequate funding, personnel, and time. Interestingly, the study found that practitioners generally lack proficiency in data science and artificial intelligence, despite these technologies being increasingly integrated into forensic tools. This gap in knowledge is concerning, given the large volumes of data investigators encounter during CSAM cases. The research also explored practitioners' perceptions of false positive and negative rates in detection technologies. Participants indicated a preference for lower rates in both categories, with a particular emphasis on reducing false positives.

As regards public opinion, Maxwell et al. [25] examined expectations regarding online child protection measures by surveying 482 Australian adults. The results indicated strong support for increased action by online service providers (OSPs) against child sexual exploitation and for government regulation of OSP responsibilities. However, respondents also expressed concerns about privacy for innocent users and data security practices of both governments and technology companies. The findings suggested that while there may be significant public backing for measures to combat online child sexual exploitation, both governments and OSPs must consider user perspectives when developing policies to address these sensitive issues.

A qualitative, in-depth study of public opinion by Deldari et al. [26] investigated users' perceptions of online CSAM detection mechanisms employed by electronic service providers (ESPs). The study focused on two types of mechanisms: server-side implementations used by companies like Google and Microsoft, and device-side implementations used by Apple. Through semi-structured interviews with 23 US-based participants who use both Apple products and at least one other ESP, the researchers explored awareness, attitudes, and privacy concerns related to these detection mechanisms. The findings revealed that most participants were initially unaware of CSAM detection mechanisms. Upon learning about them, participants generally expressed positive attitudes, recognizing their importance in combating online child abuse. However, they also voiced various privacy concerns. Many participants held misconceptions about how CSAM detection works, often overestimating the extent of data access by ESps. The study identified a lack of

transparency from ESPs regarding the existence and data practices of these mechanisms as a significant concern. Participants also expressed worries about potential misuse of these mechanisms for purposes beyond CSAM detection. When comparing server-side and device-side mechanisms, about half of the participants preferred server-side implementations, perceiving them as less invasive to privacy. Interestingly, while the majority of participants were willing to sacrifice some privacy for the social benefit of preventing CSAM distribution, some still harbored reservations about this trade-off.

However, public support for detection measures cannot be taken for granted. As Budeanu et al. [27] showed, European perceptions of artificial intelligence vary across age, education, and prior exposure. Optimism about AI's potential is often tempered by concerns over accountability and trust in governing institutions. Therefore, acceptance of intrusive AI-driven detection tools depends on the credibility and transparency of the institutions implementing them. These varied public perceptions can be understood through the concept of folk theories, which are the intuitive, informal frameworks individuals develop to make sense of and react to complex technological systems. Research on public discourse about AI ethics shows that users form distinct narratives about technology's potential for manipulation, its moral limitations, and the profit motives of tech companies, which in turn shape their ethical judgments [28]. In the context of cybersecurity, such folk theories help explain how the public navigates the trade-offs between abstract principles like privacy and concrete emotional issues like child safety, even without a full technical understanding of the mechanisms involved.

This dynamic is supported by research into public mistrust of AI in other contexts. Studies of conspiracy discourse circulating online show that anxieties are often less about a hypothetical "rogue AI" and more about how unaccountable human actors, such as corporations or state agencies, will use opaque "black box" technologies to consolidate power and control populations. The fear is not that the machine will think for itself, but that it will be used to "do the thinking for us" [29]. Regarding the CSAM debate, public apprehension may stem not only from the detection technology itself but from uncertainty about who wields it, how data is governed, and the lack of transparency in its deployment. Moreover, public attitudes towards technological governance are not static and can be deeply divided. Research analyzing online debates about replacing politicians with AI reveals a diachronic shift from early techno-optimism to more recent sociotechnical skepticism, as the initial hopes of users were tempered by real-world experiences with generative AI's limitations [30]. This points to what some scholars term a "social bifurcation of reality," [31] where different communities construct parallel systems of meaning and trust around the same technology. Therefore, the challenge of addressing online CSAM presents a tension between privacy rights and child protection imperatives. Mann et al. [32] show how, in the Australian context, claims for collective security frequently eclipse individual privacy rights in digital policy debates, with privacy often portrayed as a barrier to effective enforcement even when safeguards are weak.

As technological advancements create new avenues for CSAM distribution, policymakers and tech companies face the difficult task of developing effective detection and prevention mechanisms without unduly infringing on user privacy. This dilemma is further complicated by the global nature of the internet, which necessitates international cooperation in legislation and enforcement, yet is hampered by inconsistent laws across jurisdictions [10]. Public opinion generally supports stronger measures against CSAM, but concerns persist about data security and potential misuse of detection technologies [19]. The lack of transparency from electronic service providers regarding their CSAM detection practices exacerbates these concerns. Moreover, the rapid evolution of technology often outpaces existing laws and detection methods [33], creating a constant need for updated approaches.

The drive for automated detection tools is largely a response to the practical realities of digital forensics, where the exponential growth in data from seized devices has created a "volume challenge", leading to enormous backlogs that overwhelm traditional manual analysis [34]. However, automating the detection of harms like grooming is a complicated task. Psychologically, grooming is a manipulative process that is difficult to detect, particularly in its earliest stages [35], because many of its behaviors mimic normal social interactions [36]. Researchers have therefore focused on identifying quantifiable "red flag" behaviors [36] and developing risk scales based on factors like relationship intimacy and secrecy to enable earlier detection [35]. This difficulty in defining and moderating subjective harm is not unique to CSAM. This is a broader challenge in governing emergent online social spaces, such as social virtual reality, where the feeling of embodiment can make harassment feel more intense [37].

Translating this psychological complexity into effective AI remains a contested field of research. On one hand, some studies show that modern deep learning models like BERT have achieved very high accuracy in detecting predatory conversations on specific datasets [38]. On the other hand, different research shows that even advanced automated methods are not yet on par with human expert analysis [39], and recent tests of popular LLMs found them to be inconsistent, unreliable, and even capable of producing harmful, victim-blaming advice in grooming scenarios [40]. This technical uncertainty is compounded by a fierce legal and policy debate. Client-side scanning has been discussed as a possible way of applying detection obligations to encrypted services, but it remains an unproven approach with unresolved uncertainties [11,41]. Security experts and civil rights groups argue that mandating such systems creates "keys under doormats" [42] and introduces a "slippery slope" toward mass surveillance that could harm other vulnerable populations [43]. This approach is seen by some as a direct conflict with fundamental rights to privacy and data protection under the EU's own legal framework [44]. In contrast, some government technical experts maintain that it may be possible to engineer balanced solutions that protect both safety and privacy, though such systems remain largely theoretical [45].

1.2. The European Commission report for Flash Eurobarometer 532

The European Commission's Flash Eurobarometer 532 [46] was conducted in July 2023, on a representative sample for EU27, in order to evaluate public attitudes concerning proposed legislation that mandates online platforms to take action to identify online CSAM. The Eurobarometer report on the protection of children against online child sexual abuse reveals widespread concern across the European Union about the risks children face online. An overwhelming majority of respondents (92%) agree that children are increasingly at risk in digital environments [46]. This concern translates into strong support (78%) for proposed EU legislation that

would oblige online service providers to prevent and detect child sexual abuse on their platforms [46].

The report indicates a clear prioritization of child protection over privacy concerns. The report also notes that a striking 96% of respondents believe that detecting child abuse is more important than or equally important to online privacy rights. This sentiment is reflected in the high levels of support (85-89%) for using various technological tools to detect child sexual abuse material online, even if these tools may impact user privacy [46]. Furthermore, 87% of respondents support detecting abuse in messages and chats, with 83% supporting such detection even in end-to-end encrypted communications [46].

However, the report reveals significant variability in opinions across different demographic groups and countries. Age is a key factor influencing attitudes, with older respondents generally more supportive of detection measures and more likely to prioritize child protection over privacy concerns. In contrast, younger respondents, particularly those aged 18-24, show somewhat less support for these measures and more concern over privacy. Education level also plays a role, with lower-educated respondents more likely to prioritize child protection over privacy. Gender differences are also observable, with women more likely than men to perceive child abuse as widespread and to support reporting it to the police [46].

The report also highlights variations between countries. Support for detection measures ranges widely, from 65% in Cyprus to 83% in Czechia and Luxembourg for the proposed EU law. Similarly, perceptions of the prevalence of child abuse vary significantly, with as few as 37% in Latvia seeing it as widespread, compared to 86% in Greece. Eastern European countries generally show lower support for detection measures, pointing to regional differences [46].

Table 1

Formulation of the questionnaire items used in analysis. Source: Flash EB 532 questionnaire.

Survey question	Response option
Q1. "How widespread do you think the problem of online child sexual abuse is in your country?"	1. Very widespread 2. Fairly widespread 3. Rather rare 4. Very rare
Q2. "Which one, if any, of the following statements comes closest to your view?"	1. "The ability to detect child abuse is more important than the right to online privacy." 2. "The right to online privacy and the ability to detect child abuse are both equally important." 3. "The right to online privacy is more important than the ability to detect child abuse."
Q3. "To what extent do you agree or disagree with the following statements?"	1. Strongly agree 2. Rather agree 3. Rather disagree 4. Strongly disagree
1. Children can safely use the internet without being exposed to harmful content. 2. Children can safely use the internet without being approached by adults seeking to harm them. 3. Children are increasingly at risk online. 4. Parents, in general, know what their children are doing online. 5. Parents are under increasing pressure to make sure that their children stay safe online. 6. Tools like parental controls are not sufficient to keep children safe online 7. Online service providers (e.g. social media platforms) can play an important role in the fight against online child sexual abuse."	
Q7 "On 3 August 2024, the EU law that allows online service providers to voluntarily detect and report online child sexual abuse will expire. The EU has proposed a new law that would oblige online service providers to prevent child sexual abuse from happening on their services. If prevention fails, and in case of significant risk of child sexual abuse, the service provider could be temporarily obliged to detect and report online child sexual abuse. If this new legislation is not agreed upon by August 2024, the detection of online child sexual abuse in online exchanges – voluntary or not – becomes illegal. With that in mind, to what extent would you support or oppose this proposed law?"	1. Strongly support 2. Tend to support 3. Tend to oppose 4. Strongly oppose
Q8. "Online services providers can use various tools, including a combination of automated technology tools and human oversight, to detect child sexual abuse online. These tools may interfere to a different extent with the privacy of the users. To what extent would you support or oppose the detection of child sexual abuse by online service providers via ...	1. Strongly support 2. Tend to support 3. Tend to oppose 4. Strongly oppose
1. Tools that automatically detect images and videos of child sexual abuse material already known to the police, to identify where these images and videos are shared online again. 2. Tools based on artificial intelligence (AI) detecting new sexual abuse material images and videos shared online. 3. Tools based on artificial intelligence (AI) detecting grooming and/or imminent abuse."	
Q9. "Last year, service providers sent 1.5 million reports of child sexual abuse online stemming from the EU. Over one million of these reports - 70 per cent – came from online messages, email and chat, the rest from other sources, such as social media and gaming platforms. To what extent would you support or oppose the following?"	1. Strongly support 2. Tend to support 3. Tend to oppose 4. Strongly oppose
1. Service providers detecting child sexual abuse material and grooming conversations shared publicly (e.g. on forums, gaming platforms). 2. Service providers detecting child sexual abuse material and grooming conversations in messages (e.g. e-mail, chat) in case of a significant risk of child sexual abuse on a specific platform. 3. Service providers detecting child sexual abuse material and grooming in messages using end-to-end encryption (end-to-end encryption prevents material or conversations from being viewed by anyone other than the sender and recipient) in case of a significant risk of child sexual abuse on a specific platform."	

The Eurobarometer survey on child protection against online sexual abuse, while providing valuable insights, exhibits several limitations in its framing and methodology that may have influenced the results. In our assessment, the survey presents a simplified dichotomy between child protection and privacy rights, without adequately discussing the implications of invasive detection technologies, such as the effects of surveillance on free speech or the potential for authoritarian abuse. This omission may have led respondents to underestimate the broader social implications of the proposed measures. The questions relating to child abuse are framed in a manner that could elicit strong emotional responses, and that may have biased respondents towards supporting more invasive measures without fully considering their broader implications or potential drawbacks.

The survey lacks technical details about the proposed detection technologies, including their functioning, accuracy rates, and potential for false positives. This absence of technical context may have resulted in opinions based on incomplete information, potentially skewing the results towards uncritical support. Moreover, the survey does not present or inquire about alternative approaches to child protection that might be less invasive to privacy. This limitation in the range of options presented may have artificially constrained respondents' consideration of possible solutions.

Despite the extensive literature on CSAM detection, AI interventions, and public perceptions of online child protection, several gaps remain. Prior research often relies on small or non-representative samples, lacks cross-national coverage, or focuses narrowly on technical detection performance without considering public opinion dynamics. Moreover, studies of public attitudes tend to report aggregate levels of support or concern, overlooking latent heterogeneity in how different segments of the population weigh privacy against child protection. These limitations constrain our understanding of nuanced opinion patterns and the social acceptability of intrusive detection technologies. To address these gaps, the present study employs K-means cluster analysis on the Eurobarometer 532 dataset to identify distinct opinion profiles across the EU27. This approach allows us to move beyond mean-level reporting and reveal underlying heterogeneity in public attitudes, providing insights into how demographic, cultural, and perceptual factors shape trade-offs between privacy and child protection. Thus, we aim to answer the following research questions: To what extent do public attitudes in the European Union reveal tensions or trade-offs between privacy concerns and support for measures to combat CSAM? How does classifying EU citizens based on their attitudes towards online privacy and child protection reveal perspectives that are not apparent in aggregate public opinion data on CSAM detection measures?

2. Materials and methods

We conducted secondary data analysis of the Flash Eurobarometer 532, published in July 2023, on the topic "Protection of children against online sexual abuse." The data were collected between June 28, 2023 and July 5, 2023. The Flash Eurobarometer 532 was the most recent dataset on Protection of Children against Online Sexual Abuse. The data were collected in the context of a new regulation proposed by the European Commission aiming to prevent and combat child sexual abuse. Therefore, the limitation to July 2023 is a way of providing insightful data tailored to a specific social and legal context. We accessed the dataset through the GESIS ZACAT portal [47]. The survey was administered online (computer-assisted web interviewing) to citizens aged 18 and over. To ensure representativeness of the EU27 population aged 18 and over, we applied appropriate case weighting available in the dataset.

Our analysis incorporated variables measuring the perceived prevalence of online child sexual abuse, the relative importance of online privacy rights versus child abuse detection capabilities, perceptions of children's online experiences, and attitudes towards potential roles of OSP in detecting and managing online child sexual abuse. Additional information regarding the variables included in the analysis is available in the table below.

We rescaled variables corresponding to Q1, Q3, Q8, and Q9 so that higher codes indicated stronger agreement or support; Q7 is reported in its original coding (1 = "Strongly support", 4 = "Strongly oppose") in Tables 6 and 10 (see Table 1 for all items and their original coding). Rescaled items are named, for example, q3_1_rec, q3_2_rec etc. Factor analysis, using principal components extraction, was performed on the Q3 scale items, revealing two factors: "children are safe online" (accounting for 31% of total variance) and "society should take action to protect children online" (accounting for 23% of total variance). The results of the factor analysis for Q3 items are presented in Tables 2 and 3.

Similarly, principal component factor analysis of Q8 and Q9 items identified a single factor: "OSP should take action in many forms and situations" (accounting for 65% of total variance). Details concerning factor analysis on Q8 and Q9 items are available in Tables 4 and 5. Based on these factor analyses, we computed three summative scores by aggregating the items corresponding to the identified factor scores.

Table 2

Factor analysis for Q3 items – Total variance explained. Source: Authors' analysis of Flash EB 532.

Component	Initial Eigenvalues			Extraction Sums of Squared Loadings		
	Total	% of Variance	Cumulative %	Total	% of Variance	Cumulative %
1	2.181	31.164	31.164	2.181	31.164	31.164
2	1.58	22.577	53.741	1.58	22.577	53.741
3	0.8	11.422	65.163			
4	0.76	10.851	76.014			
5	0.702	10.027	86.041			
6	0.602	8.604	94.646			
7	0.375	5.354	100			

Table 3

Factor analysis for Q3 items – Factor loadings. Source: Authors' analysis of Flash EB 532.

Survey item:	Component	
	1	2
q3_1_rec	0.843	−0.018
q3_2_rec	0.836	−0.038
q3_3_rec	−0.207	0.703
q3_4_rec	0.684	0.039
q3_5_rec	0.156	0.648
q3_6_rec	−0.141	0.662
q3_7_rec	0.095	0.643

To classify respondents based on their opinions regarding child protection against online sexual abuse, we employed K-means cluster analysis searching for 3 and then for 4 clusters, setting the maximum number of iterations to 100. The factor analyses provided the basis for the variables used in the clustering procedure. The three summative scores computed from the factor analyses were employed as input variables in the K-means cluster analysis to classify respondents into distinct opinion groups. We finally selected a 3-cluster solution based on the distinctiveness of identified respondent types and on higher cluster interpretability. Before clustering, each classification variable was divided by its theoretical maximum possible score. This maximum-score normalization established a common upper bound of 1 but, unlike min-max normalization, did not subtract the theoretical minimum. The resulting ranges were 0.25–1.00 for Q1 and the indices derived from four-point items, and 0.33–1.00 for the three-category Q2 item. Discriminant analysis was subsequently conducted to check the internal separability of the K-means cluster classification. Because this discriminant analysis was based on the same summative scores that defined the clusters, it should be understood as a test of the internal consistency and separability of the cluster solution rather than as an independent external validation.

Missing data were handled through pairwise deletion in the factor analyses and listwise deletion in the K-means cluster and discriminant analyses. Pairwise deletion was retained for the factor analyses to maximize the information available for estimating correlations among variables. For the cluster analysis, only respondents with valid values on all clustering variables were included, ensuring that all cases were classified on the basis of the same dimensions. Descriptive statistics were calculated using all valid responses available for each variable, with missing responses excluded from the relevant denominator. Table 6 reports a variable-specific valid N. Missingness ranged from 1.3% to 8.3% across variables. As a sensitivity analysis, we also repeated the K-means procedure using pairwise deletion. The corresponding cluster centers differed by no more than 0.01 between the two analyses, while the relative cluster sizes were also highly similar. These results indicate that the identified cluster structure was robust to the treatment of missing data.

The analyses were performed with the application of the case weights provided by Eurobarometer to ensure representativeness of the EU27 population. Weighting adjusts prevalence estimates and cluster sizes, but it may also influence the variance of estimates. This limitation should be considered when interpreting the precision of our results.

3. Results

Respondents show significant concern about online child sexual abuse, with a mean score of 2.97 out of 4 indicating it is perceived as moderately widespread. Furthermore, there is strong support for detection measures, with mean scores of 3.54 for automatic detection of known material and 3.40 for AI-based detection of new material. More descriptive information for the variables included in the analysis is presented in Table 6.

3.1. The tension between privacy and child protection concerning CSAM

The three scenarios presented to EU27 respondents regarding the use of AI in combating online CSAM in questions Q8.1, Q8.2, and Q8.3 involve different levels of privacy risks and stakes. In the first scenario, where hashing technology is used to detect known CSAM material, the privacy risks are relatively low. This approach involves matching content against a database of already identified illegal

Table 4

Factor analysis for Q8 and Q9 items – Total variance explained. Source: Authors' analysis of Flash EB 532.

Component	Initial Eigenvalues		
	Total	% of Variance	Cumulative %
1	3.876	65	64.593
2	0.755	13	77.170
3	0.483	8	85.224
4	0.385	6	91.648
5	0.280	5	96.315
6	0.221	4	100

Table 5
Factor analysis for Q8 and Q9 items – Factor loadings. Source: Authors' analysis of Flash EB 532.

Survey item:	Component
	1
q8_1_rec	0.775
q8_2_rec	0.813
q8_3_rec	0.814
q9_1_rec	0.794
q9_2_rec	0.833
q9_3_rec	0.792

Table 6
Descriptive values for items included in analysis. Source: Authors' analysis of Flash EB 532.

Indicators	N	N of "Don't know" responses	Min.	Max.	Mean	Std. Deviation
Q1. "How widespread do you think the problem of online child sexual abuse is in your country"? (1 = "Very rare", 4 = "Very widespread")	24575	1691	1	4	2.97	0.72
Q2. "Which one, if any, of the following statements comes closest to your view?"	25613	652	1	3	1.41	0.53
1. The ability to detect child abuse is more important than the right to online privacy.						
2. The right to online privacy and the ability to detect child abuse are both equally important						
3. The right to online privacy is more important than the ability to detect child abuse".						
Q3.1 "Children can safely use the internet without being exposed to harmful content" (1 = "Strongly disagree", 4 = "Strongly agree")	25695	571	1	4	1.80	0.89
Q3.2 "Children can safely use the internet without being approached by adults seeking to harm them" (1 = "Strongly disagree", 4 = "Strongly agree")	25416	850	1	4	1.86	0.90
Q3.3 "Children are increasingly at risk online" (1 = "Strongly disagree", 4 = "Strongly agree")	25918	347	1	4	3.44	0.67
Q3.4 "Parents, in general, know what their children are doing online" (1 = "Strongly disagree", 4 = "Strongly agree")	25738	528	1	4	1.84	0.79
Q3.5 "Parents are under increasing pressure to make sure that their children stay safe online" (1 = "Strongly disagree", 4 = "Strongly agree")	25340	925	1	4	3.14	0.79
Q3.6 "Tools like parental controls are not sufficient to keep children safe online" (1 = "Strongly disagree", 4 = "Strongly agree")	25233	1032	1	4	3.15	0.73
Q3.7 "Online service providers (e.g. social media platforms) can play an important role in the fight against online child sexual abuse" (1 = "Strongly disagree", 4 = "Strongly agree")	25597	668	1	4	3.36	0.71
Q7 [...] "With that in mind, to what extent would you support or oppose this proposed law"? (1 = "Strongly support", 4 = "Strongly oppose")	23849	2417	1	4	1.61	0.92
Q8.1 "To what extent would you support or oppose the detection of child sexual abuse by online service providers via tools that automatically detect images and videos of child sexual abuse material already known to the police, to identify where these images and videos are shared online again"? (1 = "Strongly oppose", 4 = "Strongly support")	25257	1008	1	4	3.54	0.70
Q8.2 "To what extent would you support or oppose the detection of child sexual abuse by online service providers via tools based on artificial intelligence (AI) detecting new sexual abuse material images and videos shared online"? (1 = "Strongly oppose", 4 = "Strongly support")	24779	1487	1	4	3.40	0.78
Q8.3 "To what extent would you support or oppose the detection of child sexual abuse by online service providers via tools based on artificial intelligence (AI) detecting grooming and/or imminent abuse"? (1 = "Strongly oppose", 4 = "Strongly support")	24614	1652	1	4	3.43	0.76
Q9.1 "To what extent would you support or oppose the following: Service providers detecting child sexual abuse material and grooming conversations shared publicly (e.g. on forums, gaming platforms)"? (1 = "Strongly oppose", 4 = "Strongly support")	24905	1360	1	4	3.52	0.71
Q9.2 "To what extent would you support or oppose the following: Service providers detecting child sexual abuse material and grooming conversations in messages (e.g. e-mail, chat) in case of a significant risk of child sexual abuse on a specific platform"? (1 = "Strongly oppose", 4 = "Strongly support")	24881	1384	1	4	3.52	0.70
Q9.3 "To what extent would you support or oppose the following: Service providers detecting child sexual abuse material and grooming in messages using end-to-end encryption (end-to-end encryption prevents material or conversations from being viewed by anyone other than the sender and recipient) in case of a significant risk of child sexual abuse on a specific platform"? (1 = "Strongly oppose", 4 = "Strongly support")	24095	2171	1	4	3.47	0.75

material, which minimizes the chances of false positives and reduces the potential for infringing on innocent users' privacy. The second scenario, involving AI detection of new CSAM material, presents higher privacy risks. As the AI system would be identifying previously unknown content, there is an increased potential for false positives. This could lead to innocent content being flagged, potentially violating users' privacy and causing undue distress or harm to individuals wrongly identified. The third scenario, detecting grooming and imminent abuse, carries the highest privacy stakes. This approach requires analyzing personal communications and interactions, which is inherently more invasive than examining shared images or videos. The AI would need to interpret context and nuance in conversations, raising concerns about the extent of surveillance and the potential for misinterpretation of innocent interactions.

We expect respondents to answer differently to these questions due to the varying levels of privacy invasion and the perceived trade-offs between privacy and child protection in each scenario. The first scenario, with its focus on known material, is likely to be seen as the least controversial and most acceptable. The third scenario, while potentially the most effective in preventing abuse, might be perceived as the most intrusive, leading to more hesitation among respondents. Thus, the similarity in respondents' answers across the three scenarios, particularly for the detection of new CSAM material and grooming/imminent abuse, as observable in Fig. 1, is noteworthy and unexpected given the different privacy implications of each approach.

The similarity is even higher when examining public opinion for the questions Q9.1, Q9.2, and Q9.3. The scenarios presented for CSAM detection across different communication channels - public platforms, messages such as email and chat, and end-to-end encrypted messages - involve, again, varying levels of privacy implications and risks. For public platforms, the privacy stakes are relatively lower. Content shared publicly is already accessible to a wide audience, and users typically have reduced expectations of privacy in these spaces. Messages such as email and chat present higher privacy risks. These are often considered more personal forms of communication, where users expect a greater degree of confidentiality. Scanning these messages for CSAM could potentially expose private conversations, personal information, or sensitive data unrelated to child abuse. Lastly, end-to-end encrypted messages carry the highest privacy stakes. These systems are designed to ensure that only the sender and intended recipient can read the messages, providing a strong guarantee of privacy and security. Detecting content in an end-to-end encrypted service would require analysis at an endpoint, before encryption, after decryption, or through another mechanism outside the encrypted transmission itself. Although such an approach may leave the encryption protocol formally intact, it can weaken the practical confidentiality and security protections associated with end-to-end encryption [11].

Again, we would expect respondents to answer differently to these scenarios due to these varying privacy implications. The public platform scenario might receive the highest support, as it involves the least invasion of personal privacy, while the end-to-end encryption scenario would likely receive the least support, as it involves fundamentally altering a system designed to protect user privacy. The striking similarity in responses across the three scenarios - public platforms, messages (email, chat), and end-to-end encrypted messages - is indeed surprising given the significant differences in privacy implications (see Fig. 2).

In order to study in more depth the nuances of public opinion beyond the general consensus presented in the Eurobarometer report, we employed K-means cluster analysis to classify respondents. While the overall results show high levels of support for CSAM detection measures across various scenarios, this uniformity may mask important underlying patterns and divergences in public attitudes. K-means cluster analysis enables us to identify subgroups within the population that share similar response patterns across multiple criterion variables, potentially uncovering more perspectives on the balance between child protection and privacy concerns.

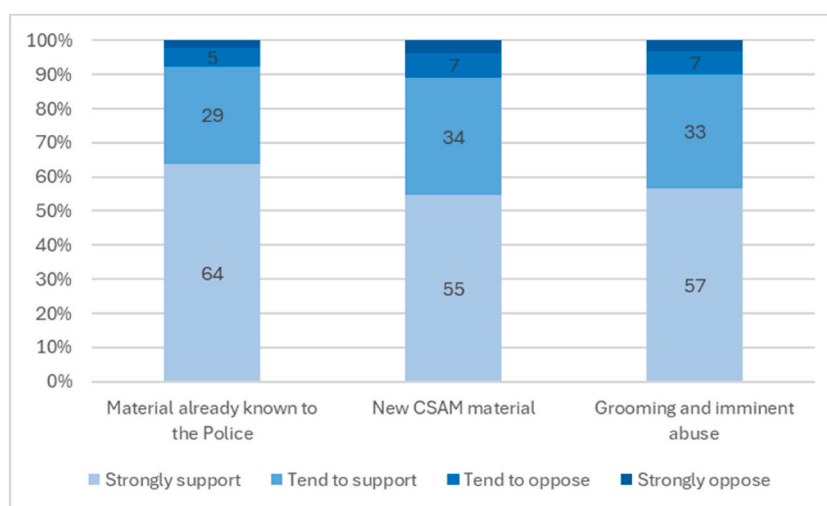


Fig. 1. Distribution of opinions concerning scenarios of using automated detection of material already known to the police, AI-based detection of new CSAM, and AI-based detection of grooming or imminent abuse. "Don't know" responses are excluded from the percentages. Source: Authors' analysis of Flash EB 532 data.

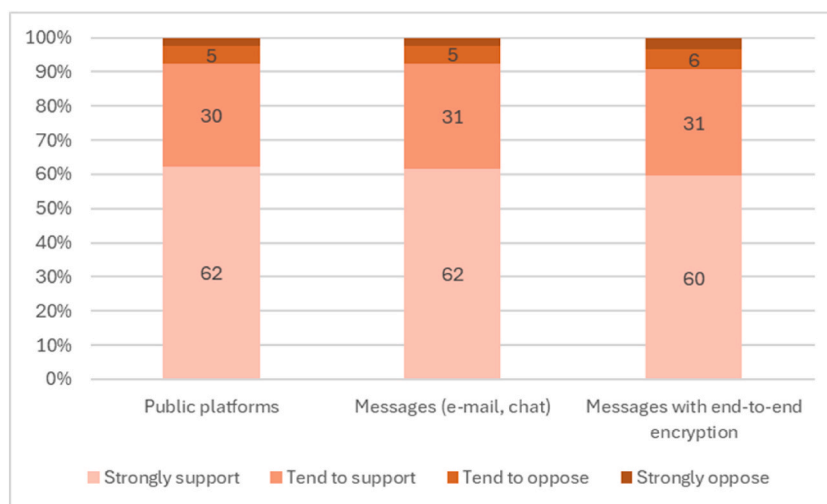


Fig. 2. Distribution of opinions concerning gradients of privacy in combatting online CSAM. “Don't know” responses are excluded from the percentages. Source: Authors' analysis of Flash EB 532 data.

3.2. Classification through cluster analysis

The cluster analysis of Flash Eurobarometer 532 respondents from EU27 reveals three distinct groups with varying perspectives on the balance between online privacy and actions against CSAM. These groups, which we labeled as “Privacy-focused”, “Action-focused”, and “CSAM-focused”, demonstrate different attitudes towards the prevalence of online child sexual abuse, the importance of online privacy, perceptions of children's online safety, and support for protective measures. The results of the K-means cluster analysis are available in Table 7.

The discriminant analysis results, available in Table 8, provide strong validation for the three-cluster solution identified through the K-means cluster analysis of EU27 respondents' opinions on online privacy and actions against CSAM. The classification results show a high accuracy rate, with 98.8% of original grouped cases correctly classified.

The “Privacy-focused” cluster exhibits a relatively high concern about the prevalence of online child sexual abuse while placing the strongest emphasis on the right to online privacy. This group also shows the highest belief that children are safe online, which may explain their more moderate stance on social and OSP involvement in child protection efforts.

In contrast, the “Action-focused” cluster demonstrates a different perception of online child sexual abuse prevalence. They view online privacy and child abuse detection as equally important and do not emphasize a tradeoff between the two. Despite having a lower belief in children's online safety, this group strongly supports both social action and OSP involvement in protecting children online.

The “CSAM-focused” cluster stands out with the highest concern about online child sexual abuse prevalence. They place the least importance on online privacy relative to child abuse detection and have the lowest belief in children's online safety. Consequently, this group shows the strongest support for both social action and OSP involvement in protecting children online.

An inverse relationship is visible between the perceived importance of online privacy and support for action against CSAM. The CSAM-focused cluster prioritizes child protection over privacy concerns, while the Privacy-focused cluster maintains a stronger emphasis on privacy rights. Additionally, the perception of how widespread online child sexual abuse is correlates with support for action, with the CSAM-focused cluster perceiving the highest prevalence and showing the strongest support for intervention. Interestingly, the Action-focused cluster, while not perceiving the highest prevalence of abuse, still strongly supports intervention,

Table 7

Results of cluster classification of respondents in three types. Source: Authors' analysis of Flash EB 532.

Indicators	Privacy-focused	Action-focused	CSAM-focused
How widespread is online child sexual abuse? (Maximum-score normalized; 1 = Very widespread)	0.72	0.59	0.84
The right to online privacy is more important than the ability to detect child abuse. (Maximum-score normalized; 1 = “The right to online privacy is more important than the ability to detect child abuse”)	0.58	0.56	0.36
Index: Children are safe online. (Maximum-score normalized; 1 = “Strongly agree” to all indicators)	0.51	0.45	0.42
Index: Society should take action to protect children online. (Maximum-score normalized; 1 = “Strongly agree” for all indicators)	0.69	0.80	0.84
Index: OSP should take action in many forms and situations. (Maximum-score normalized; 1 = “Strongly support” for all indicators)	0.55	0.89	0.92

Note: Classification variables were divided by their theoretical maximum possible scores. The resulting ranges are 0.25-1.00 for Q1 and the indices derived from four-point items, and 0.33-1.00 for the three-category Q2 item.

Table 8

Discriminant analysis for the three identified types. Source: Authors' analysis of Flash EB 532.

Cluster classification			Predicted Group Membership			Total
			Privacy-focused	Action-focused	CSAM-focused	
Original	Count	1.00 Privacy-focused	4780	42	65	4888
		2.00 Action-focused	80	7296	0	7376
		3.00 CSAM-focused	110	0	11562	11672
	%	1.00 Privacy-focused	97.8	0.9	1.3	100.0
		2.00 Action-focused	1.1	98.9	0.0	100.0
		3.00 CSAM-focused	0.9	0.0	99.1	100.0

98.8% of original grouped cases correctly classified. Note: totals are based on weighted counts; because weighted counts are rounded to whole numbers for presentation, the displayed cell values may not sum exactly to the reported row and column totals.

especially by OSP. This suggests that perceived risk is not the sole factor driving support for action.

The weighted proportions of the three clusters identified in the EU27 population (see [Table 9](#)) provide valuable insights into public opinion regarding the balance between online privacy and actions against CSAM.

The CSAM-focused cluster emerges as the largest group, comprising about 48% of the EU27 population. This indicates that nearly half of the population exhibits high concern about the prevalence of online child sexual abuse and strongly supports both social and OSP interventions to protect children online. This group prioritizes child protection over privacy concerns, suggesting widespread readiness for robust measures against CSAM. The Action-focused cluster represents 30% of the population. This substantial minority supports strong measures for both online privacy and child abuse detection, without framing a tradeoff between them. Their perspective suggests a nuanced approach to the issue, recognizing the importance of both privacy and child protection. The Privacy-focused cluster, while the smallest of the three, still represents a significant 22% of the EU27 population. This group's emphasis on online privacy rights, coupled with their support for interventions, highlights the persistent importance of privacy concerns for a considerable portion of the population. [Table 10](#) reveals differences in the approval rates of the proposed EU law to tackle online CSAM across the three identified population types.

As expected, the CSAM-focused group shows the strongest support for the proposed law. An overwhelming 73% strongly support the measure, with an additional 15% tending to support it. The Action-focused group also shows strong support, though slightly less than the CSAM-focused group. 63% strongly support the law, and 26% tend to support it. The Privacy-focused group, also as expected, shows the most reservations about the proposed law. While a majority still supports the measure (76% in total), the support is less enthusiastic compared to the other groups. 47% tend to support the law, while only 29% strongly support it. This group also shows the highest level of opposition, with 24% expressing some level of opposition.

Interestingly, across all groups, the percentage of those who strongly oppose the law is relatively consistent (ranging from 7% to 9%), suggesting a core group of opponents regardless of overall privacy or protection priorities.

To further explore the nuances masked by the aggregate data presented earlier, [Table 11](#) breaks down the mean support for specific detection scenarios across the three identified clusters. This analysis reveals the significant variations in opinion that underpin the overall trends. The results validate the cluster profiles: the CSAM-focused group displays the highest level of support on every measure, followed closely by the Action-focused group. Both of these groups show high approval for all proposed actions, regardless of the privacy implications of the technology or communication channel involved. In contrast, the Privacy-focused cluster demonstrates sensitivity to these trade-offs. While still generally supportive, their approval is lower overall and varies according to the invasiveness of the measure. For instance, their support is weakest for scanning end-to-end encrypted messages and using AI to detect grooming, with scores hovering near the neutral midpoint of the scale. Their support is highest for the least invasive action: detecting already known CSAM using hashing technology. This breakdown reveals that the apparent uniformity of public opinion is largely driven by the two majority clusters. It highlights that a substantial minority of the European public is more hesitant and more discerning about the balance between security and privacy.

3.2.1. Socio-demographic profile of population types

The socio-demographic profiles of the three clusters (Privacy-focused, Action-focused, and CSAM-focused) reveal variations across different demographic characteristics, all with statistically significant Chi-Square test results ($p < 0.001$) (see [Table 12](#)).

There is a notable gender difference in cluster membership. Women are more likely to be in the CSAM-focused cluster (54%)

Table 9

EU27 population distribution across the three identified types. Source: Authors' analysis of Flash EB 532.

		Number of cases in each cluster		% of cases in each cluster	
		Unweighted	Weighted	Unweighted	Weighted
Cluster	Privacy-focused	5346	5229	22.5%	21.8%
	Action-focused	8512	7115	35.8%	29.7%
	CSAM-focused	9916	11590	41.7%	48.4%
N		23774	23934	100%	100%
Missing cases (Listwise treatment of missing cases)		2480	2331	-	-

Table 10

Differential attitudes towards the proposed EU law across identified types. Source: Authors' analysis of Flash EB 532.

		Cluster classification			Total
		Privacy-focused	Action-focused	CSAM-focused	
Q7. [...] With that in mind, to what extent would you support or oppose this proposed law?	Strongly oppose	9%	7%	9%	8%
	Tend to oppose	15%	5%	3%	6%
	Tend to support	47%	26%	15%	25%
	Strongly support	29%	63%	73%	61%
Total		100.0%	100%	100%	100%

Table 11

Mean support for various detection scenarios across the three identified clusters. Source: Authors' analysis of Flash EB 532.

Detection scenario	Privacy-focused	Action-focused	CSAM-focused	Total
Q8.1 "To what extent would you support or oppose the detection of child sexual abuse by online service providers via tools that automatically detect images and videos of child sexual abuse material already known to the police, to identify where these images and videos are shared online again"? (1 = "Strongly oppose", 4 = "Strongly support")	2.80	3.66	3.76	3.54
Q8.2 "To what extent would you support or oppose the detection of child sexual abuse by online service providers via tools based on artificial intelligence (AI) detecting new sexual abuse material images and videos shared online"? (1 = "Strongly oppose", 4 = "Strongly support")	2.53	3.54	3.64	3.4
Q8.3 "To what extent would you support or oppose the detection of child sexual abuse by online service providers via tools based on artificial intelligence (AI) detecting grooming and/or imminent abuse"? (1 = "Strongly oppose", 4 = "Strongly support")	2.58	3.55	3.68	3.43
Q9.1 "To what extent would you support or oppose the following: Service providers detecting child sexual abuse material and grooming conversations shared publicly (e.g. on forums, gaming platforms)"? (1 = "Strongly oppose", 4 = "Strongly support")	2.76	3.65	3.74	3.52
Q9.2 "To what extent would you support or oppose the following: Service providers detecting child sexual abuse material and grooming conversations in messages (e.g. e-mail, chat) in case of a significant risk of child sexual abuse on a specific platform"? (1 = "Strongly oppose", 4 = "Strongly support")	2.70	3.63	3.75	3.52
Q9.3 "To what extent would you support or oppose the following: Service providers detecting child sexual abuse material and grooming in messages using end-to-end encryption (end-to-end encryption prevents material or conversations from being viewed by anyone other than the sender and recipient) in case of a significant risk of child sexual abuse on a specific platform"? (1 = "Strongly oppose", 4 = "Strongly support")	2.59	3.56	3.71	3.47

compared to men (42%). Conversely, men are more represented in the Privacy-focused (25%) and Action-focused (33%) clusters than women (19% and 27% respectively). This suggests that women may be more concerned about child protection online, while men tend to prioritize privacy or more nuanced approaches.

A clear age trend emerges across the clusters. Younger respondents (18-34 years) are more evenly distributed across the three clusters, with a higher representation in the Privacy-focused cluster. As age increases, there is a marked shift towards the CSAM-focused cluster. For instance, 56% of those 65 years and older are in the CSAM-focused cluster, compared to only 37% of 18-24-year-olds.

Educational attainment shows interesting patterns. Those who have never been in full-time education have a high representation in the Privacy-focused cluster (45%). Still, for those with formal education, there is a general trend of higher educational attainment being associated with greater representation in the Action-focused cluster and lower representation in the CSAM-focused cluster. This suggests that education might influence a more balanced view on the issue.

There are slight variations based on community type, with rural areas showing a slightly higher representation in the Privacy-focused cluster (24%) compared to large towns/cities (21%). However, the differences are not as pronounced as in other demographic categories.

3.2.2. Distribution of clusters across EU countries

The country-level prevalence of the three clusters across EU27 countries reveals noteworthy patterns and variations in attitudes towards online privacy and child protection (see Table 13). The visual representation of the regional distribution of population types is available in Fig. 3.

Southern European countries generally show a higher prevalence of the CSAM-focused cluster. For instance, Greece (60%), Italy (59%), and Spain (52%) have higher proportions of CSAM-focused respondents compared to the EU27 average. This suggests a stronger emphasis on child protection measures in these countries. In contrast, many Eastern European countries tend to have a higher representation in the Action-focused cluster. Bulgaria (45%), Estonia (47%), Hungary (44%), and Poland (43%) all show above-

Table 12

Socio-demographic profiles of identified population types. Source: Authors' analysis of Flash EB 532.

		Cluster membership			Statistical sig. of Chi Square tests ^a
		Privacy-focused	Action-focused	CSAM-focused	
		Row N %	Row N %	Row N %	
Gender	"Male"	25%	33%	42%	p < 0.001
	"Female"	19%	27%	54%	
Age	18 - 24 years	35%	29%	37%	p < 0.001
	25 - 34 years	31%	33%	37%	
	35 - 44 years	25%	31%	44%	
	45 - 54 years	19%	31%	51%	
	55 - 64 years	16%	30%	54%	
	65 years and older	16%	27%	56%	p < 0.001
Age at graduation	Never been in full-time education	45%	15%	40%	
	Up to 14 years	22%	21%	57%	
	15 years	20%	19%	61%	
	16 years	19%	22%	59%	
	17 years	20%	26%	54%	
	18 years	19%	30%	51%	
	19 years	19%	27%	53%	
	20 years	20%	32%	47%	
	21 years	20%	31%	49%	
	22 years and older	20%	34%	46%	
	Still in full-time education	30%	32%	39%	
Type of community	A rural area or village	24%	28%	48%	p < 0.001
	Small or medium-sized town	22%	31%	48%	
	Large town/city	21%	30%	49%	

Note: Percentages and reported Ns are weighted. Percentages may not sum to exactly 100% because of rounding.

^a Pearson and Likelihood Ratio Chi-Square Tests are statistically significant at p < 0.001.**Table 13**

Country-level prevalence of identified types. Source: Authors' analysis of Flash EB 532.

Country	Privacy-focused	Action-focused	CSAM-focused	Total
	Row %	Row %	Row %	Row %
Austria	21%	32%	47%	100%
Belgium	26%	29%	45%	100%
Bulgaria	23%	45%	32%	100%
Croatia	22%	32%	46%	100%
Cyprus (Republic)	15%	33%	52%	100%
Czech Republic	34%	37%	30%	100%
Denmark	26%	49%	25%	100%
Estonia	23%	47%	30%	100%
Finland	21%	38%	41%	100%
France	22%	29%	49%	100%
Germany	23%	23%	53%	100%
Greece	17%	23%	60%	100%
Hungary	27%	44%	29%	100%
Ireland	22%	35%	44%	100%
Italy	20%	21%	59%	100%
Latvia	31%	47%	22%	100%
Lithuania	25%	36%	40%	100%
Luxembourg	16%	42%	42%	100%
Malta	13%	52%	35%	100%
Netherlands	26%	20%	53%	100%
Poland	23%	43%	34%	100%
Portugal	12%	38%	50%	100%
Romania	13%	41%	45%	100%
Slovakia	30%	43%	27%	100%
Slovenia	24%	38%	38%	100%
Spain	20%	28%	52%	100%
Sweden	18%	32%	50%	100%

average proportions in this category.

Northern and Western European countries present more varied patterns. Denmark stands out with a notably high proportion of Action-focused respondents (49%) and one of the lowest proportions of CSAM-focused respondents (25%). The Netherlands shows a

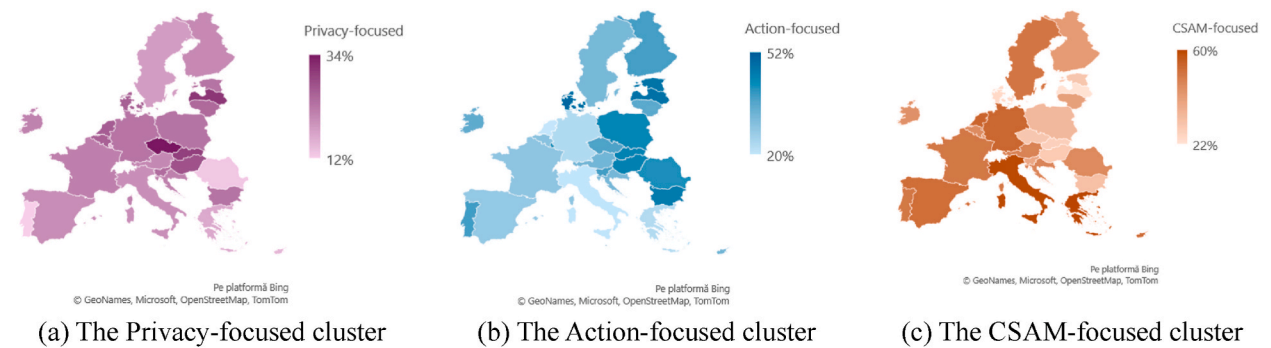


Fig. 3. Heatmaps displaying the regional variability of the identified population types across EU27 countries. Each panel maps the country-level prevalence of one cluster: panel (a) the Privacy-focused cluster, panel (b) the Action-focused cluster, and panel (c) the CSAM-focused cluster. Source: Authors' analysis of Flash EB 532.

polarized distribution, with high proportions in both the Privacy-focused (26%) and CSAM-focused (53%) clusters, but a low proportion in the Action-focused cluster (20%). Germany presents an interesting case with a high proportion of CSAM-focused respondents (53%) and equal, lower proportions of Privacy-focused and Action-focused respondents (23% each).

Some notable outliers include Czech Republic, with the highest proportion of Privacy-focused respondents (34%) and one of the lowest CSAM-focused proportions (30%). Malta has the highest proportion of Action-focused respondents (52%), followed by Denmark (49%), while Latvia and Estonia each have 47%. Latvia has the lowest CSAM-focused proportion (22%). Regional patterns are not entirely consistent. For example, while many Eastern European countries show high Action-focused proportions, Romania breaks this pattern with a high CSAM-focused proportion (45%).

4. Discussion

The Flash Eurobarometer 532 results illustrate the tension between privacy and online child protection. The survey responses demonstrate that when confronted with the emotionally charged issue of child sexual abuse, respondents tend to prioritize child protection over privacy concerns, even in scenarios that pose significant risks to user privacy. This is most striking in the similar levels of support for CSAM detection in public platforms, private messages, and end-to-end encrypted communications. The latter, in particular, represents a fundamental compromise of privacy safeguards, yet receives nearly identical support as less invasive measures.

This uniform support across scenarios suggests that respondents may not fully grasp the nuanced privacy implications of different detection methods. It appears that the emotive nature of child protection overrides more abstract concerns about data privacy and security. This process aligns with the formation of “folk theories” [28] where the public develops intuitive frameworks to make sense of complex technologies. This supports the findings of Deldari et al. [26], who discovered that most participants in their study held misconceptions about how CSAM detection technologies work. Our results suggest this phenomenon is widespread, as the lack of differentiation in support for scanning public versus encrypted platforms points to a public opinion shaped more by moral urgency than by technical understanding.

However, the apparent consensus revealed by aggregate statistics masks underlying complexities in public opinion. This is where the application of cluster analysis proves valuable. By identifying distinct groups within the population - Privacy-focused, Action-focused, and CSAM-focused - the analysis reveals the nuanced perspectives that are obscured in overall averages. The existence of these clusters demonstrates that while the tension between privacy and child protection is prevalent, it is not universal. The Privacy-focused cluster maintains a stronger emphasis on privacy rights even in the face of child protection concerns. This group's existence suggests that a significant minority does consider privacy implications more carefully, even in emotionally charged contexts. The Action-focused cluster, navigating the trade-off between privacy and child protection concerns, further illustrates the diversity of public opinion.

The emergence of a CSAM-focused cluster as the largest group (48% of the EU27 population) aligns with the strong public support for increased action against online child sexual exploitation reported by Maxwell et al. [25]. This group's willingness to prioritize child protection over privacy concerns suggests a readiness for robust measures against CSAM, which could potentially facilitate the implementation of more stringent detection and prevention mechanisms. However, the existence of a significant Privacy-focused cluster (22% of the population) provides quantitative weight to the qualitative concerns identified by Deldari et al. [26]. Their research found that even supportive users harbored reservations about the potential for misuse and the lack of transparency from service providers. Our Privacy-focused cluster represents the segment of the European public for whom these reservations are paramount.

The educational attainment patterns suggest that education might also influence a more nuanced view on the issue, possibly guided by a better familiarity with the personal risks and social implications of large-scale digital surveillance. This pattern is consistent with the findings of Budeanu et al. [27], who showed that European perceptions of artificial intelligence more broadly are stratified by age and education. Our study thus demonstrates that these same socio-demographic cleavages shape public opinion specifically in the

high-stakes context of CSAM detection, where optimism about technology is tempered by concerns over trust and accountability.

The age-related trends in cluster membership, with younger respondents more evenly distributed across clusters and older respondents predominantly in the CSAM-focused cluster, raise interesting questions about generational differences in privacy attitudes and risk perception. These findings could be interpreted in the light of the socialization process of younger individuals. Growing up in a digital age provided more diverse experiences of privacy infringements, which leads to distinctive views on privacy and security trade-offs.

The cluster analysis results also relate to the findings of Singh and Nambiar [23] regarding the potential of AI in preventing and detecting online child sexual abuse. The strong support for action among the CSAM-focused and Action-focused clusters suggests that there might be public backing for the development and implementation of AI-driven interventions. However, the privacy concerns of the Privacy-focused cluster underscore the need for careful consideration of ethical implications and potential misuse of these technologies, as highlighted in the literature.

The identification of these distinct opinion clusters is crucial for policymakers and stakeholders. It highlights the need for more nuanced approaches to public communication and policy development. While the majority may support strong child protection measures regardless of privacy implications, there is a significant portion of the population that requires more balanced approaches or stronger privacy safeguards. Moreover, the cluster analysis reveals important demographic and geographic variations in attitudes towards privacy and child protection. These variations are important for tailoring policies and communication strategies to different segments of the population or different regions within the EU.

5. Conclusion

In this study, we examined the attitudes of the EU27 population towards the detection of CSAM, focusing on the tension between privacy rights and child protection imperatives. By employing cluster analysis of the Flash Eurobarometer 532 dataset, we moved beyond aggregate measures of support and revealed three distinct perspectives in the EU27 population: a Privacy-focused group (22%), an Action-focused group (30%), and a CSAM-focused group (48%). These clusters demonstrate that, while public opinion overall strongly favors robust detection measures, there are variations in how citizens balance the values of privacy and safety.

Even when scenarios involve significant intrusions into personal communications, such as client-side scanning or detection in encrypted channels, most respondents expressed support for surveillance-oriented solutions if they were framed as protecting children. This outcome suggests that emotional salience and moral urgency can override abstract concerns about privacy, particularly in debates involving children's safety. At the same time, the existence of a sizable Privacy-focused group indicates that concerns about surveillance, data protection, and misuse of technology remain relevant and must not be neglected in policy design.

Theoretically, this study contributes to research on public perceptions of technology governance by showing how citizens organize their views on privacy and security into coherent patterns of meaning when confronted with controversial and emotionally charged policy issues. Our results support the notion that people rely on intuitive “folk theories” when evaluating complex technological interventions, prioritizing moral urgency over technical distinctions. By identifying patterned opinion profiles rather than relying on averages, our study advances a more sociologically nuanced understanding of how publics make sense of cybersecurity and surveillance measures.

The implications for policymakers and service providers are twofold. First, public communication strategies must be sensitive to the diversity of views across demographic and regional lines, acknowledging that universal measures may not resonate equally across the EU. Second, regulatory frameworks should strive for balance. Strong mechanisms to detect and prevent CSAM are needed, but their legitimacy depends on transparent safeguards that protect the fundamental rights of citizens. Addressing gaps in public understanding of detection technologies, improving transparency from electronic service providers, and ensuring independent oversight are crucial steps towards fostering trust and effectiveness of such policies.

Our study also has some limitations. We used secondary survey data whose framing may have influenced responses, particularly by emphasizing child protection without providing detailed information about the technical and legal implications of detection technologies. In addition, survey responses cannot fully capture the depth and reasoning behind individual views. Furthermore, cluster analysis imposes discrete categories on what may in reality be continuous and overlapping attitudes, potentially simplifying the complexity of public opinion. In addition, the maximum-score normalization used for clustering equalized the upper bounds of the classification variables but retained different nonzero lower bounds for four-category and three-category inputs; the cluster solution should therefore be interpreted as conditional on this scaling choice.

Future research could employ mixed-methods approaches, combining quantitative surveys with qualitative interviews to gain deeper insights into the nuances of public opinion on this issue. Additionally, longitudinal studies could track how these attitudes evolve over time, especially in response to technological advancements, policy changes, and high-profile CSAM cases. Furthermore, cross-cultural comparisons beyond the EU could provide valuable insights into how different social norms and legal frameworks influence public attitudes towards the privacy-security balance in combating online CSAM.

CRedit authorship contribution statement

Simona-Nicoleta Vulpe: Writing – review & editing, Writing – original draft, Visualization, Validation, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Răzvan Rughiniș:** Writing – review & editing, Writing – original draft, Validation, Supervision, Resources, Methodology, Investigation, Funding acquisition, Formal analysis, Conceptualization. **Dinu Țurcanu:** Writing – review & editing, Writing – original draft, Validation, Supervision,

Resources, Methodology, Investigation, Funding acquisition, Formal analysis, Conceptualization. **Daniel Rosner:** Writing – review & editing, Writing – original draft, Validation, Methodology, Investigation, Formal analysis, Conceptualization.

Data availability statement

Data are publicly available at https://search.gesis.org/research_data/ZA8763.

Ethics statement

This study is a secondary analysis of the Flash Eurobarometer 532 dataset, a publicly available and fully anonymized dataset collected by the European Commission and distributed by GESIS. The original survey was conducted in accordance with the ethical and data-protection standards governing Eurobarometer surveys. Because the present study relies exclusively on pre-existing, anonymized secondary data and involved no interaction with human participants and no access to personally identifiable information, no additional ethics approval or informed consent was required for this secondary analysis. The original data source is cited in the reference list [47] and is available through the GESIS data archive, as indicated in the Data availability statement.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work the authors used ChatGPT 5.5 for literature and language editing. The authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

References

- [1] Statista, CSAM Content Reported Online 2022, Statista, 2026. <https://www.statista.com/statistics/1448957/pieces-csam-content-reported-online-platforms/>. (Accessed 13 July 2026).
- [2] Internet Safety 101, Online sexual exploitation and abuse of children (OSEAC) 101 statistics, internet safety, 101, <https://internetsafety101.org/statistics/online-child-exploitation-abuse-statistics/>, 2025. (Accessed 13 July 2026).
- [3] Reuters, Police Take down “Kidflix” Child Abuse Platform, Europol says, Reuters, 2025. <https://www.reuters.com/world/europe/police-take-down-kidflix-child-abuse-platform-europol-says-2025-04-02/>. (Accessed 13 July 2026).
- [4] R. Rughiniş, C. Rughiniş, E. Bran, Generative AI and social engines of hate, in: J. Liebowitz (Ed.), *Regulating Hate Speech Created by Generative AI*, Auerbach Publications, Boca Raton, FL, 2024, pp. 1–18, <https://doi.org/10.1201/9781032654829-1>.
- [5] S. Ali, S.A. Pasha, A. Cox, E. Youssef, Examining the short and long-term impacts of child sexual abuse: a review study, *SN Soc. Sci.* 4 (2024) 56, <https://doi.org/10.1007/s43545-024-00852-6>.
- [6] P.M. Hellevik, L.-E.A. Haugen, C. Överlien, Outcomes of image-based sexual abuse among young people: a systematic review, *Front. Psychol.* 16 (2025) 1599087, <https://doi.org/10.3389/fpsyg.2025.1599087>.
- [7] F. Schmidt, F. Varese, S. Bucci, Understanding the prolonged impact of online sexual abuse occurring in childhood, *Front. Psychol.* 14 (2023) 1281996, <https://doi.org/10.3389/fpsyg.2023.1281996>.
- [8] European Commission, Proposal for a regulation of the european parliament and of the council laying down rules to prevent and combat child sexual abuse, COM(2022) 209 final, 2022/0155(COD), Brussels, 11 May 2022. <https://eur-lex.europa.eu/legal-content/EN/TXT/?uri=CELEX:52022PC0209>. (Accessed 13 July 2026).
- [9] H.-E. Lee, T. Ermakova, V. Ververis, B. Fabian, Detecting child sexual abuse material: a comprehensive survey, *Forensic Sci. Int. Digit. Investig.* 34 (2020) 301022, <https://doi.org/10.1016/j.fsidi.2020.301022>.
- [10] K. Parti, J. Szabó, The legal challenges of realistic and AI-driven child sexual abuse material: regulatory and enforcement perspectives in Europe, *Laws* 13 (2024) 67, <https://doi.org/10.3390/laws13060067>.
- [11] H. Abelson, R. Anderson, S.M. Bellovin, J. Benaloh, M. Blaze, J. Callas, W. Diffie, S. Landau, P.G. Neumann, R.L. Rivest, J.I. Schiller, B. Schneier, V. Teague, C. Troncoso, Bugs in our pockets: the risks of client-side scanning, *J. Cybersecur.* 10 (2024), <https://doi.org/10.1093/cybsec/tyad020> tyad020.
- [12] F. Silva, R.R. Silva, J. Bernardino, A survey of generative AI for detecting pedophilia crimes, *Appl. Sci.* 15 (2025) 7105, <https://doi.org/10.3390/app15137105>.
- [13] K. McQue, AI is overpowering efforts to catch child predators, experts warn, *Guardian* (2024). <https://www.theguardian.com/technology/article/2024/jul/18/ai-generated-images-child-predators>. (Accessed 13 July 2026).
- [14] A. Krishna, Internet.gov: tech companies as government agents and the future of the fight against child sexual abuse, *Calif. Law Rev.* 109 (2021) 1581–1635, <https://doi.org/10.15779/Z38KW57J9B>.
- [15] D. Milmo, Chatbot site depicting child sexual abuse images raises fears over misuse of AI, *Guardian* (2025). <https://www.theguardian.com/technology/2025/sep/21/chatbot-site-depicting-child-sexual-abuse-images-raises-fears-over-misuse-of-ai>. (Accessed 13 July 2026).
- [16] A. Pinget, Controversial proposal on combating child sexual abuse online, *euclid*. <https://euclid.eu/news/proposal-on-combating-child-sexual-abuse-online/>, 2022. (Accessed 13 July 2026).
- [17] A. Hern, Planned EU rules to protect children online are attack on privacy, warn critics, *Guardian* (2022). <https://www.theguardian.com/society/2022/may/12/planned-eu-rules-to-protect-children-online-are-attack-on-privacy-warn-critics>. (Accessed 13 July 2026).

- [18] M.G. Jones, Planned EU laws on child sexual abuse have sparked a privacy row. Why? Euronews (2023). <http://www.euronews.com/my-europe/2023/10/19/planned-eu-laws-on-child-sexual-abuse-have-sparked-a-bitter-privacy-row-why>. (Accessed 13 July 2026).
- [19] R. Rughiniş, S.-N. Vulpe, D. Ţurcanu, D. Rosner, Balancing act: europeans' privacy calculus and security concerns in online CSAM detection, *Front. Big Data* 8 (2025) 1477911, <https://doi.org/10.3389/fdata.2025.1477911>.
- [20] I. Atrey, Cybercrime and its legal implications: analysing the challenges and legal frameworks surrounding cybercrime, including issues related to jurisdiction, privacy, and digital evidence, *Int. J. Res. Anal. Rev.* 10 (2023). <https://ssrn.com/abstract=4789133>.
- [21] INHOPE - Association of Internet Hotline Providers, Global CSAM Legislative Overview 2024, INHOPE, 2024. <https://inhope.org/EN/articles/inhope-global-csam-legislative-overview-2024>. (Accessed 13 July 2026).
- [22] A. Razi, S. Kim, A. Alsoubai, G. Stringhini, T. Solorio, M. De Choudhury, P.J. Wisniewski, A human-centered systematic literature review of the computational approaches for online sexual risk detection, *Proc. ACM Hum.-Comput. Interact.* 5 (CSCW2) (2021) 1–38, <https://doi.org/10.1145/3479609>, 465.
- [23] S. Singh, V. Nambiar, Role of artificial intelligence in the prevention of online child sexual abuse: a systematic review of literature, *J. Appl. Secur. Res.* 19 (2024) 586–627, <https://doi.org/10.1080/19361610.2024.2331885>.
- [24] L. Sanchez, C. Grajeda, I.M. Baggili, C. Hall, A practitioner survey exploring the value of forensic tools, AI, filtering, and safer presentation for investigating child sexual abuse material (CSAM), *Digit. Investig.* 29 (2019) S124–S142, <https://doi.org/10.1016/j.diin.2019.04.005>.
- [25] F. Maxwell, M. Salter, N. Peleg, “To say report it, well, it seems a little useless”: evaluating Australians' expectations of online service providers and reducing online child sexual exploitation, *Policy Internet* 16 (2024) 384–410, <https://doi.org/10.1002/poi3.378>.
- [26] E. Deldari, P. Thakkar, Y. Yao, Users' perceptions of online child abuse detection mechanisms, *Proc. ACM Hum.-Comput. Interact.* 8 (CSCW1) (2024) 1–26, <https://doi.org/10.1145/3637424>, 147.
- [27] A.-M. Budeanu, D. Ţurcanu, D. Rosner, European perceptions of artificial intelligence and their social variability: an exploratory study, in: 2023 24th International Conference on Control Systems and Computer Science (CSCS), IEEE, Bucharest, Romania, 2023, pp. 436–443, <https://doi.org/10.1109/CSCS59211.2023.00075>.
- [28] D.M. Obreja, D. Rosner, R. Siminiuc, Folk theories of ethical agency on Reddit threads: negotiating morality in AI data assemblages, in: 2025 24th Roedunet Conference: Networking in Education and Research (Roedunet), IEEE, Chişinău, Moldova, 2025, pp. 1–6, <https://doi.org/10.1109/RoEduNet68395.2025.11208366>.
- [29] A.M.A. Dobra, D.A. Iordache, C. Rughiniş, A. Corcaci, Computational analysis of AI mistrust: generative thematic mapping of conspiracy discourse on Reddit, in: 2025 24th Roedunet Conference: Networking in Education and Research (Roedunet), IEEE, Chişinău, Moldova, 2025, pp. 1–6, <https://doi.org/10.1109/RoEduNet68395.2025.11208432>.
- [30] C. Ghiţuleasa, D.A. Iordache, A. Corcaci, Computational discourse analysis of AI-as-governor debates: a decade of Reddit discussions on replacing politicians, in: 2025 24th Roedunet Conference: Networking in Education and Research (Roedunet), IEEE, Chişinău, Moldova, 2025, pp. 1–6, <https://doi.org/10.1109/RoEduNet68395.2025.11208462>.
- [31] C. Rughiniş, M.G. Flaherty, The social bifurcation of reality: symmetrical construction of knowledge in science-trusting and science-distrusting discourses, *Front. Sociol.* 7 (2022) 782851, <https://doi.org/10.3389/fsoc.2022.782851>.
- [32] M. Mann, A. Daly, M. Wilson, N. Suzor, The limits of (digital) constitutionalism: exploring the privacy-security (im)balance in Australia, *Int. Commun. Gaz.* 80 (2018) 369–384, <https://doi.org/10.1177/1748048518757141>.
- [33] O. Cullen, K.Z. Ernst, N. Dawes, W. Binford, G. Dimitropoulos, “Our laws have not caught up with the technology”: understanding challenges and facilitators in investigating and prosecuting child sexual abuse materials in the United States, *Laws* 9 (2020) 28, <https://doi.org/10.3390/laws9040028>.
- [34] D. Quick, K.-K.R. Choo, Impacts of increasing volume of digital forensic data: a survey and future research challenges, *Digit. Investig.* 11 (2014) 273–294, <https://doi.org/10.1016/j.diin.2014.09.002>.
- [35] P. Pasca, F. Signore, C. Tralci, D.D. Gottardo, M. Longo, G. Preite, E. Ciavolino, Detecting online grooming at its earliest stages: development and validation of the online grooming risk scale, *Mediterr. J. Clin. Psychol.* 10 (2022), <https://doi.org/10.13129/2282-1619/mjcp-3248>.
- [36] E.L. Jeglic, G.M. Winters, B.N. Johnson, Identification of red flag child sexual grooming behaviors, *Child Abuse Negl.* 136 (2023) 105998, <https://doi.org/10.1016/j.chiabu.2022.105998>.
- [37] L. Blackwell, N. Elliott, N. Elliott-Deflo, R. Schwartz, Harassment in social virtual reality: challenges for platform governance, *Proc. ACM Hum.-Comput. Interact.* 3 (2019) 1–25, <https://doi.org/10.1145/3359202>. CSCW) Article 100.
- [38] N. Agarwal, T. Ünlü, M.A. Wani, P. Bours, Predatory conversation detection using transfer learning approach, in: G. Nicosia, V. Ojha, E. La Malfa, G. La Malfa, P. M. Pardalos, G. Giuffrida, R. Umeton (Eds.), *Machine Learning, Optimization, and Data Science, Lecture Notes in Computer Science*, vol 13164, Springer, Cham, 2022, pp. 488–499, https://doi.org/10.1007/978-3-030-95467-3_35.
- [39] D. Cook, M. Zilka, H. DeSandre, S. Giles, A. Weller, S. Maskell, Can We Automate the Analysis of Online Child Sexual Exploitation Discourse? arXiv:2209.12320 [cs.CL], 2022 <https://doi.org/10.48550/arXiv.2209.12320>.
- [40] E. Prosser, M. Edwards, Helpful or harmful? Exploring the efficacy of large language models for online grooming prevention, in: Proceedings of the 2024 European Interdisciplinary Cybersecurity Conference (EICC 2024), Association for Computing Machinery, New York, NY, 2024, pp. 1–10, <https://doi.org/10.1145/3655693.3655694>.
- [41] P. Rosenzweig, The Law and Policy of client-side Scanning, Lawfare, 2023. <https://www.lawfaremedia.org/article/law-and-policy-client-side-scanning>. (Accessed 13 July 2026).
- [42] H. Abelson, R. Anderson, S.M. Bellovin, J. Benaloh, M. Blaze, W. Diffie, J. Gilmore, M. Green, S. Landau, P.G. Neumann, R.L. Rivest, J.I. Schiller, B. Schneier, M. A. Specter, D.J. Weitzner, Keys under doormats: mandating insecurity by requiring government access to all data and communications, *J. Cybersecur.* 1 (2015) 69–79, <https://doi.org/10.1093/cybsec/tyv009>.
- [43] T.R. Ringenberg, L.G. Kisselburgh, A way forward: what we know (or not) about CSAM and privacy, in: USENIX Conference on Privacy Engineering Practice and Respect (PEPR '22), 2022. <https://www.usenix.org/conference/pepr22/presentation/ringenberg>. (Accessed 13 July 2026).
- [44] S. Chousou, J. Magaud, L. Pavoni, M. Williams, Is encryption a fundamental right? A case study on CSAM regulation in the EU, sciences Po chair digital, in: Governance and Sovereignty, 2023, pp. 1–41. <https://www.sciencespo.fr/public/chaire-numerique/wp-content/uploads/2023/07/Encryption.pdf>. (Accessed 13 July 2026).
- [45] I. Levy, C. Robinson, Thoughts on Child Safety on Commodity Platforms, arXiv:2207.09506 [cs.CY], 2022, <https://doi.org/10.48550/arXiv.2207.09506>.
- [46] European Commission, Flash eurobarometer 532: protection of children against online sexual abuse - report. <https://europa.eu/eurobarometer/surveys/detail/2656>, July 2023. (Accessed 13 July 2026).
- [47] European Commission, Flash Eurobarometer 532: protection of Children against online sexual abuse - July 2023, ipsos european public affairs [producer], GESIS data catalogue, ZA8763, dataset version 1.0.0. <https://doi.org/10.4232/1.14214>, 2023.